complex exponentials.

Even/odd decomposition: A way of breaking a signal into two other signals, one having even symmetry, and the other having odd symmetry.

Even order filter: An analog or digital filter having an even number of poles.

False-negative: One of four possible outcomes of a target detection trial. The target is present, but incorrectly indicated to be not present.

False-positive: One of four possible outcomes of a target detection trial. The target is not present, but incorrectly indicated to be present.

Fast Fourier transform (FFT): An efficient algorithm for calculating the discrete Fourier transform (DFT). Reduces the execution time by *hundreds* in some cases.

FFT convolution: A method of convolving signals by multiplying their frequency spectra. So named because the FFT is used to efficiently move between the time and frequency domains.

Field: Interlaced television displays the even lines of each frame (image) followed by the odd lines. The even lines are called the *even field*, and the odd lines the *odd field*.

Filter kernel: The impulse response of a filter implemented by convolution. Also known as the *convolution kernel* and the *kernel*.

Filtered backprojection: A technique used in computed tomography for reconstructing an image from its views. The views are *filtered* and then *backprojected*.

Finite impulse response (FIR): An impulse response that has a finite number of nonzero values. Often used to indicate that a filter is carried out by using convolution, rather than recursion.

First difference: An operation for discrete signals that mimics the first derivative for continuous signals; also called the *discrete derivative*.

Fixed point: One of two common ways that computers store numbers; usually used to store integers. See *floating point* for comparison.

Flat-top window: A window used in spectral analysis; provides an accurate measurement of the amplitudes of the spectral components. The windowed-sinc filter kernel can be used.

Floating point: One of the two common ways that computers store numbers. Floating point uses a form of scientific notation, where a mantissa is raised to an exponent. See *fixed point* for comparison.

Forward transform: The analysis equation of the Fourier transform, calculating the frequency domain from the time domain. See *inverse transform* for comparison.

Fourier reconstruction: One of the methods used in computed tomography to calculate an image from its views.

Fourier series: The member of the Fourier transform family that deals with time domain signals that are *continuous* and *periodic*.

Fourier transform: A family of mathematical techniques based on decomposing signals into sinusoids. In the complex version, signals are decomposed into complex exponentials.

Fourier transform pair: Waveforms in the time and frequency domains that correspond to each other. For example, the rectangular pulse and the sinc function.

Fovea: A small region in the retina of the eye that is optimized for high-resolution vision.

Frame: An individual image in a television signal. The NTSC television standard uses 30 frames per second.

Frame grabber: A analog-to-digital converter used to digitize and store a frame (image) from a television signal.

Frequency domain: A signal having frequency as the independent variable. The output of the Fourier transform.

Frequency domain aliasing: Aliasing that occurs occurring in the frequency domain in response to an action taken in the time domain. Aliasing during sampling is an example.

Frequency domain convolution: Convolution carried out by multiplying the frequency spectra of the signals.

Frequency domain encoding: One of two main ways that information can be encoded in a signal. The information is contained in the amplitude, frequency, and phase of the signal's component sinusoids. Audio signals are the best example. See *time domain encoding* for comparison.